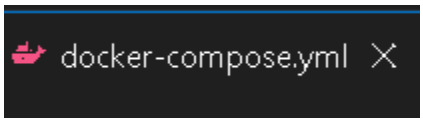
Docker Compose

If the command is to complex like:
docker run -d --env MYSQL_ROOT_PASSWORD="ghat123@gmail.com" --env MYSQL_DATABASE="employee_db" --name mysqldb_new --network my-net mysql

Other member of the team must also need to provide the same command.

So, let's create docker compose file. Which reduces the complexity of writing big commands.
Provide extension like **.yml** or **.yaml.**



services : is kind of containers/images
For the command: **docker run -d --env MYSQL_ROOT_PASSWORD="ghat123@gmail.com" --env MYSQL_DATABASE="employee_db" --name mysqldb_new --network my-net mysql**

```
services:
  ▷Run Service
  mysqldb_new:
    image: "mysql:latest"
    environment:
      - MYSQL_ROOT_PASSWORD="ghat123@gmail.com"
      - MYSQL_DATABASE="employee_db"
```

Mysqldb_new is the image/service.
Image: image
Environment Variables: MYSQL_DATABASE, MYSQL_ROOT_PASSWORD
Container_name: is optional
Initially we dont have any images nor containers.

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker images -a

IMAGE    ID              DISK USAGE   CONTENT SIZE   EXTRA
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

Folder structure:

```
PYTHON CODE
docker-compose.yml
Dockerfile
PyCode.py
```

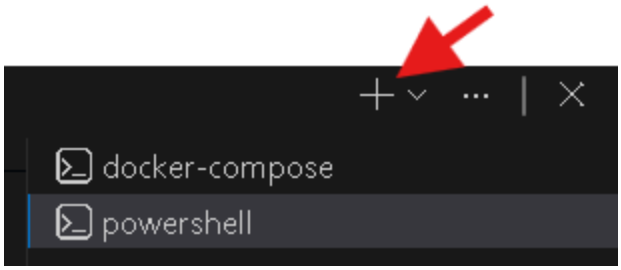
```
erfile   PyCode.py   docker-compose.yml
er-compose.yml
Run All Services
services:
  Run Service
  mysqldb:
    image: "mysql:latest"
    environment:
      - MYSQL_ROOT_PASSWORD="ghat123@gmail.com"
      - MYSQL_DATABASE="employee_db"
    container_name: "mysqldb"
```

Running the file:

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose up
[+] up 7/13
 - Image mysql:latest [⣿⣿⣿⣿ ⣿⣿⣿⣿ ⣿⣿⣿⣿⣿⣿⣿⣿] 39.88MB / 281.9MB Pulling
```

```
Consider choosing a different directory.
mysqldb  | 2026-03-24T09:47:32.484681Z 0 [Sys
mysqldb  | 2026-03-24T09:47:32.484754Z 0 [Sys
MySQL Community Server - GPL.
```

Then open the second terminal:




```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker images

IMAGE           ID             DISK USAGE   CONTENT SIZE   EXTRA
mysql:latest    24e450bbd24f       1.27GB          283MB     U
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

So docker-compose has pulled the mysql:latest image.

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS
b9cde3822bce   mysql:latest   "docker-entrypoint.s…"   4 minutes ago   Up 4 minutes
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

```
STATUS          PORTS                    NAMES
Up 4 minutes    3306/tcp, 33060/tcp      mysqldb
```

It is in running. And provided a name as mysqldb.

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose down
[+] down 2/2
 ✓ Container mysqldb           Removed
 ✓ Network pythoncode_default  Removed
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

In old terminal,

```
mysqldb  | 2026-03-24T09:53:22.258746Z 0 [System] [
6.0).
mysqldb exited with code 137
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker images

IMAGE          ID             DISK USAGE   CONTENT SIZE   EXTRA
mysql:latest   24e450bbd24f       1.27GB          283MB
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker images -a

IMAGE          ID             DISK USAGE   CONTENT SIZE   EXTRA
mysql:latest   24e450bbd24f       1.27GB          283MB
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

It automatically removed container, no need for using **–rm.**

By using detached mode:

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose up -d
[+] up 2/2
 ✓ Network pythoncode_default Created
 ✓ Container mysqldb          Started
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```